HUMAN–COMPUTER INTERACTION · DETAILED STUDY NOTES

Unit III — Designing the Screen

Graphical screen design, visible language, grids, the design principles, Shneiderman's 8 Golden Rules and Nielsen's 10 Heuristics — explained simply, with diagrams, comparison tables and viva prep.

Subject BCSED1-713 **Programme** B.Tech CSE · 7th Sem **University** MRSPTU, Bathinda

> THE ONE WORD BEHIND THE WHOLE UNIT
>
> **Usability** = how easy, efficient and satisfying a system is to use. Every idea in this unit — good layout, a grid, the golden rules, the heuristics — exists to make the interface more usable.

01

What this unit is about

Units I and II were about understanding users and evaluating designs. Unit III is the *how do I actually lay out the screen* part: how to arrange text, images, colour and space so a screen looks clear (graphical screen design), and what tested rules make any interface easy to use (the design principles, golden rules and heuristics).

Usability is usually broken into **five measurable qualities** (Jakob Nielsen):

- **Learnability** — how quickly a first-time user can do basic tasks.
- **Efficiency** — once learned, how fast users get things done.
- **Memorability** — after a break, how easily they return to competence.
- **Errors** — how few mistakes users make, and how easily they recover.
- **Satisfaction** — how pleasant the system is to use.

EXAMPLE A ticket-booking app with clear steps, easy payment and a visible confirmation has better usability than one with confusing buttons and no confirmation.

> USEFUL DISTINCTION
> **Utility** = does it have the features you need? **Usability** = are those features easy to use? A system is *useful* only when it has both.

02

Graphical screen design

> DEFINITION
> The skill of arranging the visual elements of a screen — text, images, buttons, colour and empty space — so the screen is **clear, attractive and easy to understand**. A good screen guides the eye to the most important thing first and hides nothing important.

EXAMPLE A college website shows the university name at the top, the menu below it, notices in the centre and contact details at the bottom.

Goals of good screen design

- **Clear visual hierarchy** — the most important item stands out the most.
- **Simplicity** — keep it uncluttered so the user is not overwhelmed.
- **Consistency** — every screen feels part of the same product.
- **Low effort** — reduce how much the user must read, think and do.

The seven graphical design concepts

A Balance

Spread visual "weight" evenly so the screen isn't heavier on one side.

EX A large photo on the left balanced by text on the right.

B Alignment

Line elements up along common edges or invisible lines; makes a screen look neat.

EX All labels left-aligned, all input boxes in one column.

C Proximity (grouping)

Related items close together; unrelated items apart. The eye reads groups.

EX First + last name together, away from payment.

D Contrast

Make important things look different — bigger, bolder, brighter — to catch the eye.

EX One bright "Buy Now" button among grey elements.

E Emphasis / focal point

One clear element the eye lands on first — the key action or message.

EX A landing page's big headline + single sign-up button.

F White space

Empty area around elements; gives the design room to breathe. Need not be white.

EX Apple pages: lots of space around one product image.

G Repetition / consistency

Reuse colours, fonts and spacing so the design feels like one whole.

EX Same heading style and same blue for every link.

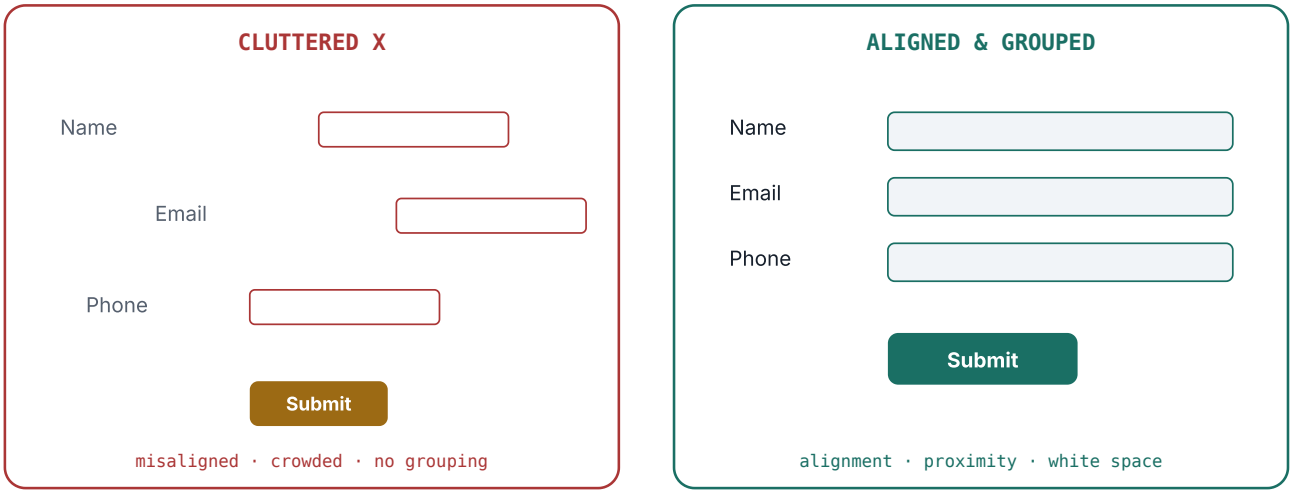


Fig 1 — the same form: cluttered & misaligned (left) vs aligned, grouped, spaced (right)

REMEMBER

Balance steadies · **alignment** tidies · **proximity** groups · **contrast** stands out · **emphasis** gives a focal point · **white space** uncrowds · **repetition** unifies.

Visible language components

DEFINITION

Visible language is everything on the screen that communicates visually — the way text, colour, shape and images carry meaning. These are the building blocks a designer picks from.

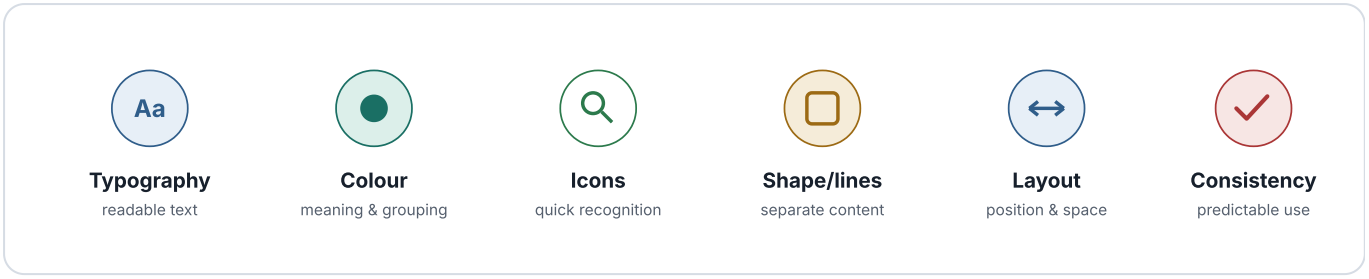


Fig 2 — the six visual building blocks that carry meaning on a screen

1 · Typography

Font, size, weight and spacing. Text carries most meaning, so it must be readable. Use large bold headings and comfortable body text, keep line length readable, and don't mix too many fonts. A clear **hierarchy** (heading > subheading > body) tells the eye what to read first.

EXAMPLE A news app uses one bold headline font and one clean article font.

2 · Colour

For meaning and grouping, not just decoration. Keep a small, consistent palette and enough **contrast** between text and background (WCAG asks for at least 4.5:1 for normal text).

EXAMPLE Red for errors, green for success, one brand colour for buttons.

MISTAKE Ten random bright colours, or light-grey text on white no one can read.

3 · Icons & imagery

Symbols that carry meaning faster than words; best paired with a text label.

EXAMPLE Magnifying glass = search · trash can = delete · house = home.

4 · Shape, lines & borders

Boxes, dividers and lines that separate and group content.

EXAMPLE A bordered card around each product on a shopping app.

5 · Layout & spacing

Where elements sit and the gaps between them — what makes a screen feel calm, not crowded.

EXAMPLE Comfortable gaps between paragraphs and around buttons.

6 · Consistency of components

The same component looks and behaves the same way across the app, so users learn it once.

EXAMPLE A banking app's "Pay" button: same green, shape and corner on every screen.

REMEMBER

Typography makes it readable · **colour** carries meaning · **icons** aid recognition · **shapes/lines** separate · **layout** organises · **consistency** makes behaviour predictable.

04

Grid-based design

DEFINITION

A **grid** is an invisible framework of columns, rows, gutters and margins used to place elements. Grid-based design means snapping all content to this framework — like writing on ruled paper instead of a blank page — so the screen looks organised, aligned and consistent.

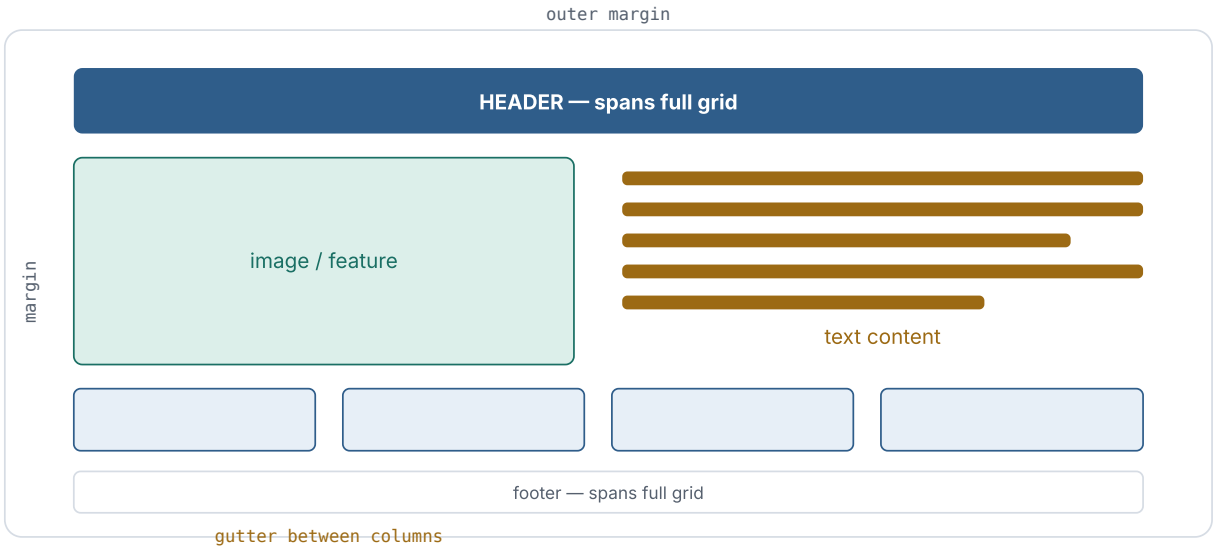


Fig 3 — a 4-column grid: margins, columns and gutters keep header, image, text and footer aligned

Parts of a grid

- **Columns** — vertical strips content sits in (sites often use 12).
- **Gutters** — gaps between columns so content doesn't touch.
- **Margins** — the empty border around the whole screen.
- **Modules / rows** — horizontal divisions; a block can span several columns.

Why grids help

- Everything **lines up** → clean, professional look.
- **Consistency** across many screens.
- Design is **faster** — snap elements to the lines.
- Supports **responsive design** — columns rearrange for phones vs laptops.

EXAMPLE A shopping site shows four product cards across on a laptop, reflowing to two across on a phone.

05

Design principles supporting usability

A screen must be **usable**, not just pretty. These principles (several from Don Norman's work) directly improve usability.

1 Simplicity

Keep only what's needed; remove clutter ("less is more").

EX A calculator shows just common buttons, not dozens of rare functions.

2 Visibility

Show options and the current state clearly; don't hide key functions behind secret gestures.

EX A music app clearly shows Play, Pause and the current song.

3 Feedback

Respond to every action so users know it worked.

EX Pressing "Send" shows "Message sent ✓".

4 Affordance

Make things look like how they're used, so users act without instructions.

EX A button looks raised/clickable; a text box looks ready to type in.

5 Mapping

Controls relate naturally to their effect; control layout matches what it controls.

EX Volume slider up = louder; stove knobs arranged like the burners.

6 Constraints

Prevent errors by only allowing valid actions.

EX "Submit" greyed out until all required fields are correctly filled.

7 Learnability & flexibility

Easy for beginners (learnability); faster paths for experts (flexibility).

EX Beginners use menus; experts use keyboard shortcuts.

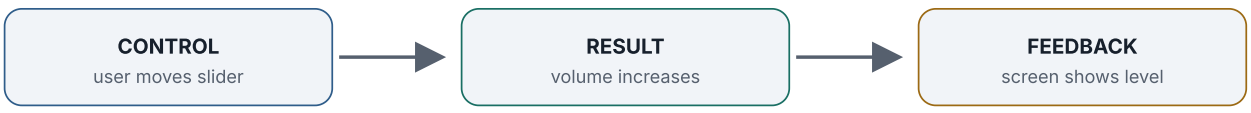


Fig 4 — a good interface links the control, its effect, and clear feedback

06 · SHNEIDERMAN

The 8 Golden Rules

Ben Shneiderman's **Eight Golden Rules** are a practical checklist used mainly *while designing*. Learn all eight, in order, with one example each.

1 Strive for consistency

Same words, colours, fonts, layouts and actions on every screen; follow known conventions.

EX "Save" is a blue button bottom-right everywhere.

MISTAKE One screen "Delete", another "Remove" for the same action.

2 Enable shortcuts for frequent users

Hotkeys, gestures or macros so experts aren't slowed by beginner steps. *In the 6th edition this rule is rephrased "Seek universal usability."*

EX Ctrl+C / Ctrl+V; double-tap to like on Instagram.

3 Offer informative feedback

Every action gets a response; small actions modest, big actions prominent.

EX "Message sent ✓"; an upload progress bar; WhatsApp's double tick.

4 Design dialogs to yield closure

Sequences with a clear beginning, middle and end; signal completion.

EX Checkout: cart → address → pay → "Order Confirmed! #1234".

5 Offer simple error handling

Stop serious mistakes; if an error happens, give a simple, clear fix without losing work. *In the 6th edition this rule is rephrased "Prevent errors."*

EX Form won't submit until valid; shows "Please enter a valid email" by the field.

6 Permit easy reversal of actions (undo)

Let users undo; reduces anxiety and encourages exploration.

EX Ctrl+Z; Gmail "Undo Send"; Restore from Trash.

7 Support internal locus of control

Users feel in charge; the system doesn't act surprisingly on its own.

EX No videos auto-playing with sound; the app asks before changes.

MISTAKE Pop-ups that hijack the screen, or settings that change by themselves.

8 Reduce short-term memory load

STM is limited (about 7±2, roughly 5–9 items). Show information instead of making users remember it.

EX Order summary on the payment page; saved addresses auto-fill; password rules shown.

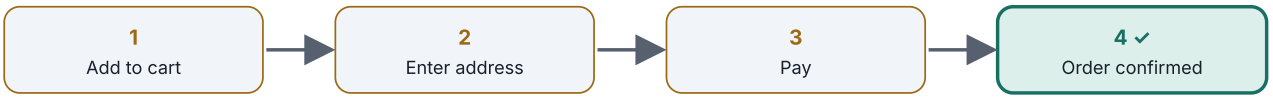


Fig 5 — a checkout sequence ends with a clear confirmation, giving closure (Rule 4)

MEMORY AID · 8 RULES

Consistency · **S**hortcuts · **F**eedback · **C**losure · **E**rror handling · **R**eversal · **C**ontrol · **M**emory load.

07 · NIELSEN

The 10 Usability Heuristics

Jakob Nielsen's **10 Heuristics** are "rules of thumb" used to *judge* an interface — the standard checklist for heuristic evaluation.

1 Visibility of system status

Keep users informed through timely feedback.

EX A loading spinner; "3 of 5 photos uploaded"; "... is typing".

MISTAKE Click "Pay", nothing happens, user clicks again and pays twice.

2 Match between system and the real world

Familiar words, icons and concepts in a natural order.

EX A trash-can icon; "Home" instead of "root directory".

MISTAKE "Error 0x8007: null pointer exception".

3 User control and freedom

A clearly marked "emergency exit", plus Undo and Redo.

EX A "Cancel" button in every dialog; a back arrow; "Undo" after archiving.

4 Consistency and standards

Same words/actions mean the same thing; follow conventions.

EX Blue underlined text = link; "X" always closes.

MISTAKE "Log out" / "Sign off" / "Exit" for the same action.

5 Error prevention

Better than a good message — stop the error happening at all.

EX A date-picker instead of typing; "Are you sure you want to delete?"

6 Recognition rather than recall

Make options visible; recognising beats recalling.

EX A dropdown of all choices; a "recently used files" list.

MISTAKE A command line forcing users to memorise exact commands.

7 Flexibility and efficiency of use

Accelerators for experts, simple for beginners.

EX Keyboard shortcuts, saved templates, favourites, custom dashboards.

8 Aesthetic and minimalist design

No irrelevant or rarely-needed information.

EX Google's homepage — just a search box.

MISTAKE A page crammed with flashing ads and walls of text.

9 **Help users recognise, diagnose & recover from errors**

Plain-language messages: what went wrong + a solution.

EX "No internet — check your Wi-Fi and try again."

10 **Help and documentation**

Ideally usable without help; when needed, easy to search and task-focused.

EX A searchable Help Centre, tooltips, a "?" icon, an FAQ.

Important differences (comparison tables)

A · Golden Rules vs Heuristics

Both aim at usability. Shneiderman's rules are guidelines used **while designing**; Nielsen's heuristics are the checklist used **while evaluating** an existing design.

Shneiderman's Golden Rules	Nielsen's Heuristics
Eight rules	Ten heuristics
By Ben Shneiderman	By Jakob Nielsen
Used mainly as design guidelines	Used mainly as an evaluation checklist
Consistency, shortcuts, feedback, closure, error handling, reversal, control, memory load	System status, real-world match, control, standards, error prevention, recognition, efficiency, minimalism, error recovery, help

B · Affordance vs Mapping vs Feedback

Concept	Main question	Example
Affordance	What can I do with this?	A button looks clickable.
Mapping	What will this control do?	Slider right increases volume.
Feedback	What happened after I acted?	A message confirms the file was saved.

C · Error prevention vs recovery

- **Prevention** — stop a mistake before it happens. *Disable Submit until valid.*
- **Recovery** — help fix it after. *Explain an invalid email and how to fix it.*

D · Recognition vs recall

- **Recognition** — pick an option shown on screen (a recent file).
- **Recall** — remember info unaided (an exact command). Good design favours recognition.

Heuristic evaluation

DEFINITION

A method where a few **experts** (not real users) inspect an interface against a list of heuristics — usually Nielsen's 10 — to find usability problems before real users do.

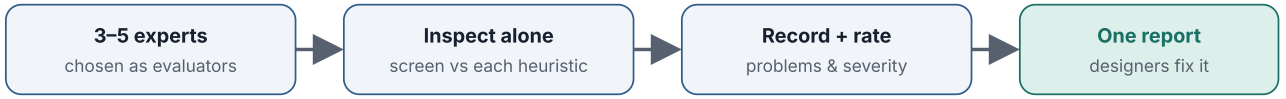


Fig 6 — the heuristic-evaluation process, from evaluators to a single fix-it report

How it is done

- Use **3–5 evaluators** — about five find roughly 75% of problems.
- Each inspects the interface **alone**, against each heuristic.
- They record problems and **rate severity**.
- Findings combine into **one report** for designers.

Advantages & limitation

- **Fast, cheap**, possible early — even on a paper prototype.
- Systematic — every screen checked against a known list.
- **Limitation:** experts aren't real users, so some real-world problems are missed.

EXAMPLE Three experts reviewing a food-delivery app note "no confirmation after ordering" (heuristic 1) and "no way to cancel an order" (heuristic 3) — both fixed before launch.

10

HCI (design) patterns

DEFINITION

A proven, **reusable solution** to a common design problem. Designers reuse patterns already known to work instead of inventing a new solution each time.

Navigation
Hamburger menu (≡), tabs or breadcrumbs.

Search
Search box with magnifying-glass icon + suggestions.

Form
Labels above fields, inline validation, one primary button.

Login / sign-up
Email + password + "forgot password".

Pagination / infinite scroll
Standard ways to show long lists.

Notification & feedback
Toasts, badges, progress bars.

WHY PATTERNS ARE USEFUL

They save design time and improve **learnability** and **consistency** because users already know them. *Example:* nearly every shopping app puts the cart icon top-right — follow the pattern, don't invent something unfamiliar.

11

Final exam revision — at a glance

Topic	What to remember
Usability	Easy, efficient, satisfying · learnability, efficiency, memorability, errors, satisfaction
Graphical screen design	Arrange text/image/colour/space clearly
Seven visual concepts	Balance, alignment, proximity, contrast, emphasis, white space, repetition
Visible language	Typography, colour, icons, shape/lines, layout, consistency
Grid-based design	Columns, rows/modules, gutters, margins → alignment + responsive
Design principles	Simplicity, visibility, feedback, affordance, mapping, constraints, learnability/flexibility
Golden Rules (8)	Consistency, shortcuts, feedback, closure, error handling, reversal, control, memory load
Heuristics (10)	Nielsen's evaluation checklist — used by 3–5 experts
Heuristic evaluation	Experts inspect vs the heuristics, rate severity, report fixes
HCI design patterns	Reusable solutions: menu, search, form, login

How to answer viva questions

For each concept remember three things: **(1) definition, (2) purpose, (3) example.** Use simple language and a real example — understanding beats word-for-word memorising.

> SAMPLE ANSWER · FEEDBACK
> "Feedback means the system responds to a user's action and tells them what happened. For example, after clicking Submit, the app shows 'Form submitted successfully'. This tells the user the action completed."

Apply the same shape to any term — e.g. *Affordance?* "An element's look tells you how to use it (definition); its purpose is to let people act without instructions; for example a raised button looks clickable."
